

BUSFIN 8200 - Problem Set 1

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September 20, 2026

Problem 1

1 (a)

Starting from $R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$ and assuming that D is greater than zero, we take log of both sides and have:

$$\begin{aligned} r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - \log(P_t) \\ &= \log(P_{t+1}) \left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - p_t \\ &= p_{t+1} + \log\left(1 + \frac{e^{d_{t+1}}}{e^{p_{t+1}}}\right) - p_t \\ &= p_{t+1} + \log(1 + e^{d_{t+1} - p_{t+1}}) - p_t \\ &= p_{t+1} + \log(1 + e^{dp_{t+1}}) - p_t \\ &= p_{t+1} - p_t + f(dp_{t+1}) \end{aligned}$$

where $f(dp_{t+1})$ is a log function of the log dividend-price ratio. Applying a first-order Taylor expansion around $\bar{dp}$ to $f(dp_{t+1})$, we have Equation (1):

$$r_{e,t+1} \approx p_{t+1} - p_t + f(\bar{dp}) + f'(\bar{dp})(dp_{t+1} - \bar{dp})$$

We define $\kappa = \frac{1}{1+e^{\bar{dp}}}$. Consequently, we have the following:

$$\begin{aligned} f'(\bar{dp}) &= \frac{e^{\bar{dp}}}{1 + e^{\bar{dp}}} = 1 - \kappa \\ e^{\bar{dp}} &= \frac{1}{\kappa} - 1 \\ \bar{dp} &= \log\left(\frac{1}{\kappa} - 1\right) \end{aligned}$$

Plugging in the above results into Equation (1), we have:

$$\begin{aligned} r_{e,t+1} &\approx p_{t+1} - p_t + \log(1 + e^{\bar{dp}}) + (1 - \kappa)dp_{t+1} - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) \\ &= p_{t+1} - p_t + \log\left(\frac{1}{\kappa}\right) - (1 - \kappa)p_{t+1} + (1 - \kappa)d_{t+1} - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) \\ &= \kappa p_{t+1} - p_t + (1 - \kappa)d_{t+1} + \kappa_0 \\ &= \kappa_0 - \kappa dp_{t+1} + (d_{t+1} - d_t) + (d_t - p_t) \\ &= \kappa_0 - \kappa dp_{t+1} + \Delta d_{t+1} + dp_t \end{aligned}$$

where $\kappa_0 = \log\left(\frac{1}{\kappa}\right) - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) = -\log(\kappa) - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right)$. Rearranging the above equation, we have:

$$\begin{aligned} dp_t &= -\kappa_0 + \kappa dp_{t+1} + r_{e,t+1} - \Delta d_{t+1} \\ \Rightarrow dp_{t+1} &= -\kappa_0 + \kappa dp_{t+2} + r_{e,t+2} - \Delta d_{t+2} \\ \Rightarrow dp_{t+2} &= -\kappa_0 + \kappa dp_{t+3} + r_{e,t+3} - \Delta d_{t+3} \end{aligned}$$

Keeping substituting these equations into the first one and generalizing the pattern H times, we have:

$$\begin{aligned}
dp_t &= -\kappa_0 \underbrace{(1 + \kappa + \kappa^2)}_{\sum_{h=1}^H \kappa^{h-1}} + \kappa^3 dp_{t+3} + [r_{e,t+1} + \kappa r_{e,t+2} + \kappa^2 r_{e,t+3}] - [\Delta d_{t+1} + \kappa \Delta d_{t+2} + \kappa^2 \Delta d_{t+3}] \\
&= -\kappa_0 \left(\frac{1 - \kappa^H}{1 - \kappa} \right) + \kappa^H \cdot dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \\
&= \kappa_0 \left(\frac{\kappa^H - 1}{1 - \kappa} \right) + \kappa^H \cdot dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \tag{1.1}
\end{aligned}$$

Given that $\mathbb{E}_t[dp_t] = dp_t$, we take conditional expectation on both sides of Equation (1.1) and have:

$$dp_t = \kappa_0 \left(\frac{\kappa^H - 1}{1 - \kappa} \right) + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \tag{1.2}$$

Since $e^{\overline{dp}}$ is always positive, we have $0 < \kappa < 1$. Therefore, as $H \rightarrow \infty$, $\kappa^H \rightarrow 0$. In addition, we also need to impose No-Bubble Condition such that $\lim_{H \rightarrow \infty} \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] = 0$. Giglio, Maggiori, and Stroebl (2016) indeed shows that there were no systemic deviations in the discounts of extremely long leaseholds to otherwise identical freeholds in the U.K. and Singapore housing markets, thus indicating no rational bubbles. Consequently, letting $H \rightarrow \infty$, Equation (1.2) can be simplified as follows:

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \tag{1.3}$$

1 (b)

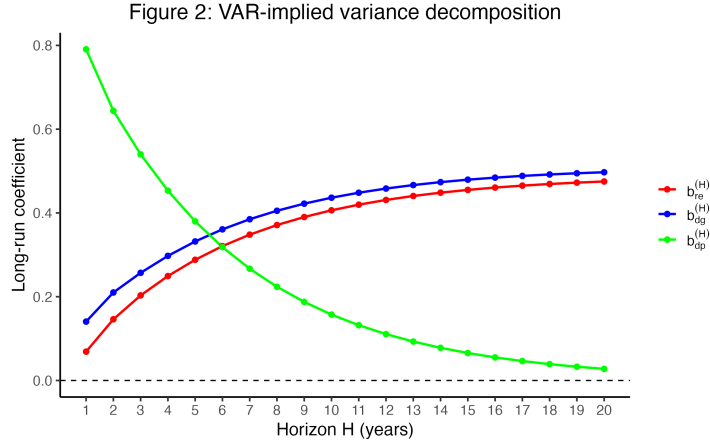
Using Equation 1.4 from the problem set, I decomposed the variation in the log dividend-price ratio into variations associated with the discounted future returns, discounted future dividend growth and terminal dividend-price ratio. By constructing these regressors at a 12-month increment, I regressed each component on dp_t in univariate OLS models to estimate their respective betas through different horizons as shown below.



The figure indicates that over the short horizon (up to 5 years), the discounted future dividend-price ratio is persistent and responsible for the majority of the variation in the log dividend-price ratio, as demonstrated by a strongly positive $\beta_{dp}^{(H)}$. Indeed, at $H = 1$, this component explains 80% of the variation in dp_t . However, as the horizon extends beyond year 1, $\beta_{dp}^{(H)}$ gradually declines and approaches zero at $H = 12$, before turning negative for the rest of the horizons. This further proves the no-bubble condition imposed in part (a). Meanwhile, from $H = 4$ onward, the discounted future return component quickly rises and becomes the dominant force in explaining the variability in dp_t , with $\beta_{re}^{(H)}$ reaching 0.9 at $H = 20$. This further supports the notion that the discount rate component is more relevant in explaining long-run, rather than short-run, variation in the log dividend-price ratio. As for the dividend growth component, $\beta_{dg}^{(H)}$ remains relatively flat, hovering around 0.2 before gradually converging to 0.1 at $H = 20$. While the log dividend-price ratio has some relationship with discounted future dividend growth, it is considerably weaker than its relationship with discounted future expected returns, especially in the long horizons. Indeed, this evidence is consistent with Cochrane's address to Stambaugh, stating that log dividend-price ratio's inability to forecast future dividend growth proves that stock returns are predictable.

1 (c)

Applying Equation 1.4 to a VAR(1) model, I constructed Figure 2 as below.



In particular, let $z_t = [dg_t, r_{e,t}, dp_t]$ follow $z_{t+1} = \Gamma_0 + \Gamma z_t + \varepsilon_{t+1}$ where $\varepsilon_{t+1} \stackrel{\text{i.i.d.}}{\sim} (0, \Sigma)$. Running these overlapping annual regressions, I recover $\hat{\Gamma}$ in the form of $(z_t' z_t)^{-1} z_t' z_{t+1}$ to forecast three state variables in the future. Then, I regress dg_t , $r_{e,t}$, and dp_t onto dp_t using univariate full-sample OLS specifications to recover their coefficients in the form of the b_z vector. As a result, I calculate the VAR-implied coefficients as follows:

$$\begin{aligned}
 b_{dg}^{(H)} &= -\mathbf{1}_{dg}' (\Gamma - \kappa^H \cdot \Gamma^{H+1}) (I - \kappa \cdot \Gamma)^{-1} b_z, \\
 b_{re}^{(H)} &= \mathbf{1}_{re}' (\Gamma - \kappa^H \cdot \Gamma^{H+1}) (I - \kappa \cdot \Gamma)^{-1} b_z, \\
 b_{dp}^{(H)} &= 1 - b_{dg}^{(H)} - b_{re}^{(H)}
 \end{aligned}$$

1 (d)

Using the same VAR process in equation (1.5) and taking H to ∞ , the term $\kappa^H \cdot dp_{t+H}$ should go to 0 under the no-bubble condition. However, there is still a residual VAR-implied b_{dp}^∞ , albeit minor at 0.0011919. This could be attributed to the first-order specifications of the VAR process. Otherwise, the VAR-implied covariance coefficients for the discount rate and dividend growth components are as follows:

Table 1: VAR-implied infinite-horizon coefficients

Component	Coefficient
$b_{re}^{(\infty)}$	0.489191
$b_{dg}^{(\infty)}$	0.509617

Table 1 suggests that under the VAR(1) specifications, eventually both expected return and dividend growth equally contribute to the variation in the log dividend-price ratio in the extremely long run.

1 (e)

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

Taking log of both sides, we have:

$$\begin{aligned}
 r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - p_t \\
 &= \log\left(P_{t+1} + \frac{P_{t+1} \cdot D_{t+1}}{P_{t+1}}\right) - p_t \\
 &= \log\left(P_{t+1} \left(1 + \frac{D_{t+1}}{P_{t+1}}\right)\right) - p_t \\
 &= p_{t+1} + \underbrace{\log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right)}_{\equiv dy_{t+1}} - p_t
 \end{aligned} \tag{1.4}$$

We also have:

$$dy_t = \log\left(1 + \frac{D_t}{P_t}\right) = \log\left(1 + \frac{e^{d_t}}{e^{p_t}}\right) = \log(1 + e^{dp_t})$$

$$\begin{aligned}
 \Rightarrow -dy_t &= -\log(1 + e^{dp_t}) = \log(1 + e^{dp_t})^{-1} \\
 \Rightarrow e^{-dy_t} &= e^{\log(1 + e^{dp_t})^{-1}} = \frac{1}{1 + e^{dp_t}} \\
 \Rightarrow \frac{1}{e^{-dy_t}} - 1 &= e^{dp_t} \\
 \Rightarrow \frac{1 - e^{-dy_t}}{e^{-dy_t}} &= e^{dp_t} \\
 \Rightarrow 1 - e^{-dy_t} &= e^{dp_t} \cdot e^{-dy_t} \\
 \Rightarrow \log(1 - e^{-dy_t}) &= dp_t - dy_t
 \end{aligned} \tag{1.5}$$

Rewriting Equation (1.4) and plugging in Equation (1.5), we have:

$$\begin{aligned}
 r_{e,t+1} - (d_{t+1} - d_t) &= (d_t - p_t) - (d_{t+1} - p_{t+1}) + dy_{t+1} \\
 \Rightarrow r_{e,t+1} - \Delta d_{t+1} &= dp_t - dp_{t+1} + dy_{t+1} \\
 &= (dp_t - dy_t) - (dp_{t+1} - dy_{t+1}) + dy_t \\
 &= \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}}) + dy_t
 \end{aligned} \tag{1.6}$$

Applying a first-order Taylor approximation to $\log(1 - e^{-dy_t})$ around $\overline{dy}$, we have:

$$\log(1 - e^{-dy_t}) = f(dy_t) \approx f(\overline{dy}) + f'(\overline{dy})(dy_t - \overline{dy}).$$

Define $\kappa \equiv e^{-\bar{dy}}$. We have the following:

$$\begin{aligned} f(\bar{dy}) &= \log(1 - e^{-\bar{dy}}) = \log(1 - \kappa), \\ f'(\bar{dy}) &= \frac{1}{1 - e^{-\bar{dy}}} \cdot e^{-\bar{dy}} = \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} = \frac{\kappa}{1 - \kappa}. \end{aligned}$$

As a result,

$$f(dy_t) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}).$$

Similarly,

$$f(dy_{t+1}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}).$$

Plugging these two equations into Equation (1.6), we have:

$$\begin{aligned} r_{e,t+1} - \Delta d_{t+1} &= \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}) - \log(1 - \kappa) - \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}) + dy_t \\ &= \frac{\kappa}{1 - \kappa}(dy_t - dy_{t+1}) + dy_t \\ &= \frac{\kappa + 1 - \kappa}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1} \\ &= \frac{1}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1} \\ &= \frac{dy_t - \kappa dy_{t+1}}{1 - \kappa}. \end{aligned}$$

Rearranging the terms, we have:

$$dy_t = (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}$$

Recursively substituting dy_{t+h} , we have:

$$dy_t = (1 - \kappa) \cdot \left[\sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \right] + \kappa^H \cdot dy_{t+H} \quad (1.7)$$

Taking the conditional expectation of both sides and assuming that $\mathbb{E}_t[X] = X$, we have:

$$dy_t = (1 - \kappa) \cdot \left[\sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \right] + \kappa^H \cdot dy_{t+H} \quad (1.8)$$

Knowing that $0 < \kappa < 1$ by construction and assuming that the no-bubble condition holds such that $\lim_{H \rightarrow \infty} \kappa^H \cdot dy_{t+H} = 0$, we let $H \rightarrow \infty$ in Equation (1.8) and have:

$$dy_t = (1 - \kappa) \cdot \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \right] \quad (1.9)$$

Problem 2

2 (a)

Running OLS regressions of average excess return on dividend-price ratio presents the following figure.



For the first few horizons, R_{adj}^2 remained low around 0.2, suggesting the dividend-price ratio's weak ability to predict average excess return in the short run. However, as the horizons extend, R_{adj}^2 gradually improves and reaches a peak of approximately 0.45 at $H = 14$. This means that over the longer horizon, current dividend-price ratio explains about 45% of the variation in the long-run average excess return, rendering this independent variable relevant in predicting long-term return.

2 (b)

I estimated the one-year OLS regression of excess return on current dividend-price ratio and retrieved the fixed estimated $\hat{b}$ and $\hat{e}$:

$$xR_{e,t+1} = a + b \cdot D_t/P_t + e_t$$

For each estimator, I calculated a corresponding spectral density matrix $\hat{S}$ before feeding it into the following sandwich covariance matrix estimator:

$$\widehat{\text{Avar}}(\hat{\theta}) = \frac{1}{T}Q^{-1}\hat{S}Q^{-1}, \quad Q = \frac{1}{T}X'X$$

Taking square root of the diagonal entries of the coefficient covariance matrix, I derive the following table:

Method	$\hat{b}$	Std. Error	t -stat
OLS	2.8038	0.3802	7.3754
White (1980)	2.8038	0.6587	4.2569
Newey–West (1987), 11 lags	2.8038	1.2990	2.1584
Hansen–Hodrick (1980), 11 lags	2.8038	1.4537	1.9287
Newey–West (1987, 1994), automatic (22 lags)	2.8038	1.2523	2.2390

As the estimate accounts for both heteroskedasticity and autocorrelation, the standard error becomes larger and the t -stat declines.

2 (c)

The Amihud and Hurvich (2004) correction responds to Stambaugh's concern for bias in the coefficient estimate due to the persistent nature of the predictor, or $\frac{D_t}{P_t}$ in this case. Indeed,

estimating the AR(1) regression of D_{t+1}/P_{t+1} on D_t/P_t such that $D_{t+1}/P_{t+1} = \theta + \phi \cdot D_t/P_t + u_{t+1}$, I retrieved $\hat{\phi} = 0.7197$, suggesting that shocks to dividend-price ratio disappear verly slowly. Given our finite sample, the OLS estimate of the persistence coefficient tends to be biased downward so that $\mathbb{E}_t[\hat{\phi} - \phi] < 0$. As a result, a downwardly biased $\hat{\phi}$ also leads to an upward bias in $\hat{b}$ via a negative covariance between shocks to return (e_{t+1}) and shocks to predictor (u_{t+1}). In order to correct the OLS's overstated return predictability, I re-estimate the slope coefficient using the Amihud and Hurvich (2004) correction as follows:

1. Step 1: Convert the dataset from months into corresponding T years. Construct the the bias-corrected slope estimate $\hat{\phi}^c = \hat{\phi} + (1/T) \cdot (1 + 3 \cdot \hat{\phi}) + (1/T^2) \cdot 3 \cdot (1 + 3 \cdot \hat{\phi})$.
2. Step 2: Subbing $\hat{\phi}^c$ into the AR(1) process, I calculate the bias-corrected residual estimate: $\hat{u}_{t+1}^c = D_{t+1}/P_{t+1} - (\hat{\theta} + \hat{\phi}^c \cdot D_t/P_t)$.
3. Step 3: I plug the updated AR(1) regression in place of D_t/P_t into Equation (2.2) in part (b) and retrieve a new $\hat{b}_{AH}$: $xR_{e,t+1} = a + b_{AH} \cdot D_t/P_t + b_u \cdot \hat{u}_{t+1}^c + e_{t+1}$.

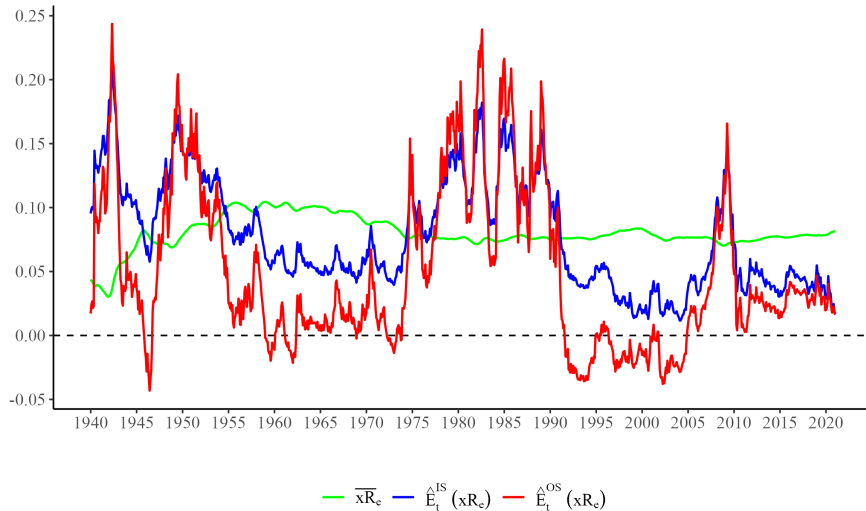
The slope estimates for the conventional OLS and Amihud and Hurvich are $\hat{b} = 2.8038$ and $\hat{b}_{AH} = 2.3366$ respectively. This empirical result is in line with the small sample bias discussion above, proving that the corrected $\hat{u}_{t+1}^c$ controls for the contemporaneous correlation between the return and dividend-price ratio innovations and shrinks the inflated slope coefficient.

2 (d)

The in-sample OLS forecast uses the full-sample data following Equation 2.2 to derive in-sample fitted values $\hat{E}_t^{IS}[xR_e]$. The out-of-sample OLS uses an expanding window with first forecast using the dividend-price ratio from December 1927 to November 1939 and excess return from December 1928 to November 1940 to fit $xR_{e,t+1} = a_t^{OS} + b_t^{OS} \cdot D_t/P_t + e_{t+1}$ and retrieve out-of-sample fitted values $\hat{E}_t^{OS}[xR_e]$. The average historical expanding-window excess return $\overline{xR}_{e,t}$ is calculated as the sample mean during the same period. Using the out-of-sample forecasting errors, I calculate the out-of-sample R_{OS}^2 as follows:

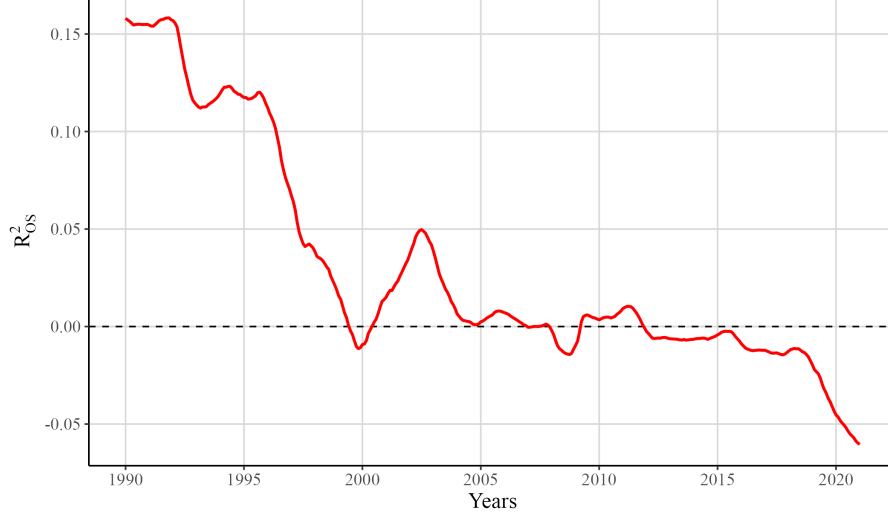
$$R_{OS}^2 = 1 - \frac{\sum_{t=1}^T \left(xR_{e,t+1} - \hat{E}_t^{OS}[xR_{e,t}] \right)^2}{\sum_{t=1}^T \left(xR_{e,t+1} - \frac{1}{T} \sum_{j=1}^T xR_{e,j+1} \right)^2} = 0.001824$$

Figure 4: Time series of sample mean, IS and OS estimates for xR



The 50-year rolling window R_{OS}^2 uses 600 observations of forecasting errors from the OS OLS estimation with the first window covering January 1941 to December 1990. As observed from the plot below, the predictability power of the model severely declines around 2000 and beyond 2005.

Figure 5: Rolling 50-year out-of-sample R-squared



2 (e)

I keep the same $\hat{E}_t^{IS}[xR_e]$ and $\overline{xR_{e,t}}$ as constructed in question 2d. For the out-of-sample estimation, assuming that dividend growth is unexpected and expected return is 100% persistent, Equation 1.9 implies:

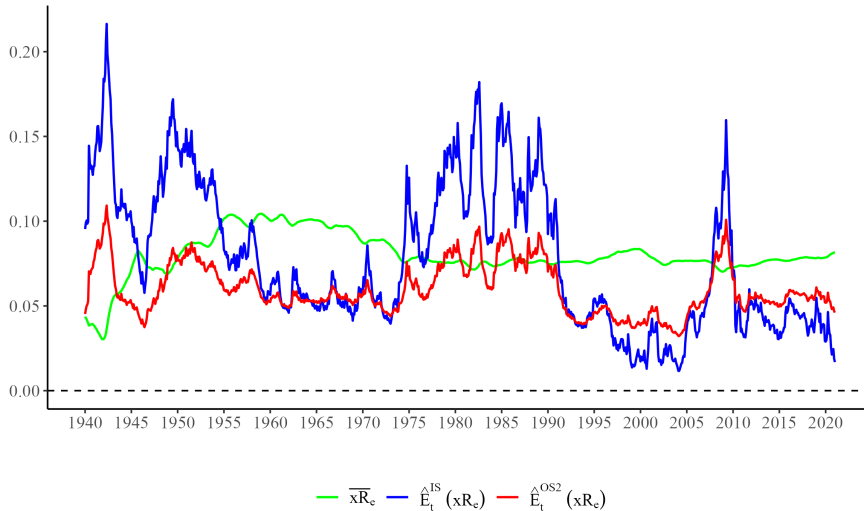
$$\log(1 + D_t/P_t) = \mathbb{E}_t[r_e] - g$$

where $\mathbb{E}_t[\Delta d_{t+h}] = g$ and $\mathbb{E}_t[\Delta r_{e,t+h}] = \mathbb{E}_t[r_e]$. Assuming 0% real interest rate, $e^{rf} = 1$. Let $G_t = \exp(g)$. Taking the exponent of both sides and subtracting e^{rf} , we have:

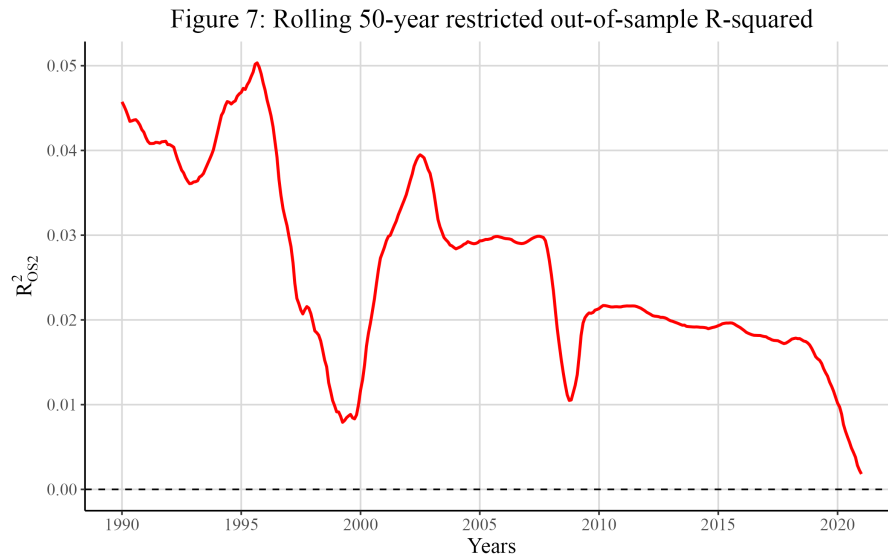
$$xR_{e,t+1} = (G_t - 1) + G_t \cdot D_t/P_t$$

Applying the steady state implied coefficients, I set $\hat{a} = \bar{G}_t - 1$ and $\hat{b} = \bar{G}_t$ such that $\bar{G}_t$ is the average $e^{\Delta d}$ over the expanding window period (starting from December 1927), I estimate $\hat{E}_t^{OS2}[xR_e]$ and retrieve the following figure.

Figure 6: Time series of sample mean, IS and restricted OS estimates



R_{OS2}^2 , using the same formula as question 2d, is 0.001978 with the 50-year rolling R_{OS2}^2 time series as follows:



Given these results, we can clearly see that the steady state constraint has slightly improved the out-of-sample R^2 ($0.001978 > 0.001824$). Looking over the horizons, the rolling R^2 using the fixed coefficient constraint also never dipped below 0, suggesting that imposing restrictions from steady-state valuation model can improve OS predictability. This makes sense as less estimation errors come at the expense of less flexibility in estimating regression coefficients. This conclusion is consistent with the view of Campbell & Thompson (2008).